Collision in Cryptographic Hash Functions

SHA-1, SHA-256, SHA-3

1. Collisions in Hash Functions

A cryptographic hash function is a mathematical algorithm that takes an input of arbitrary size and produces a fixed-length output known as a hash value. For example, a simple text such as “hello” can be transformed into a long hexadecimal string. Regardless of whether the input is a single character or a large file, the output size remains constant.

Hash functions are widely used in modern systems for password storage, data integrity verification, and digital signatures. Their security relies heavily on certain properties, one of the most important being **collision resistance**.

2. Collision and Its Necessity

A collision occurs when two different inputs produce the same hash output. Formally, for a hash function H:

$H(x_1) = H(x_2)$, where $x_1 \neq x_2$

This implies that two distinct messages become indistinguishable when only their hash values are considered.

Collisions are mathematically guaranteed due to the **pigeonhole principle**, since hash functions map an infinite input space to a finite output space (e.g., 2^256 possible outputs for a 256-bit hash). Therefore, at least two inputs must share the same hash.

However, the objective of a secure hash function is not to eliminate collisions, but to ensure that they are **computationally infeasible to find**.

3. Collision Resistance and Its Importance

Collision resistance means that it should be practically impossible to find two distinct inputs that produce the same hash. This property is critical for security.

If collisions were easy to find, an attacker could:

- Replace a legitimate file with a malicious one that has the same hash
- Forge digital signatures
- Compromise data integrity

Thus, while collisions exist in theory, they must remain unreachable in practice.

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4. Mathematical Foundation: Birthday Paradox

The complexity of finding collisions is governed by the birthday paradox. Instead of requiring 2^n attempts, collisions can be found in approximately:

N ≈ 2^(n/2)

where n is the number of bits in the hash output.

Applying this:

- SHA-1 (160-bit) → 2^80 attempts
- SHA-256 / SHA-3 (256-bit) → 2^128 attempts

These values are extremely large and form the mathematical basis of collision resistance.

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5. Types of Collision Attacks

Collision attacks can be categorized into three types:

5.1 Brute-Force Search

Random inputs are generated and hashed until a collision is found. This requires exploring the entire input space and is computationally infeasible.

5.2 Birthday Attack

Previously computed hashes are stored and compared. This reduces complexity to 2^(n/2), but still remains impractical for strong hash functions.

5.3 Cryptanalytic Attacks

These exploit weaknesses in the structure of the hash function to significantly reduce complexity. Unlike brute-force methods, they rely on mathematical insights.

A key example is the SHAttered attack on SHA-1.

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6. Computational Feasibility of Collision Attacks

To evaluate feasibility, we analyze brute-force, birthday attacks, and computational limits.

Case 1: Brute-Force Collision Search

Number of attempts:

$N \approx 2^n$

SHA-1:
$N \approx 2^{160} \approx 1.46 \times 10^{48}$

SHA-256 / SHA-3:
$N \approx 2^{256} \approx 1.16 \times 10^{77}$

Assume GPU speed:

$R = 10^9$ hashes/sec

Time:

$T = N / R$

SHA-1:
$T \approx 10^{39}$ sec $\approx 10^{31}$ years

SHA-256 / SHA-3:
$T \approx 10^{68}$ sec $\approx 10^{60}$ years

Comparison:

- Age of Earth $\approx 10^9$ years
- Age of Universe $\approx 10^{10}$ years

Thus:

$10^{60} >> 10^{10}$

Brute-force search is physically impossible.

Case 2: Birthday Attack

Number of attempts:

$N \approx 2^{(n/2)}$

SHA-1:
$N \approx 2^{80} \approx 1.2 \times 10^{24}$

SHA-256 / SHA-3:
$N \approx 2^{128} \approx 3.4 \times 10^{38}$

Time:

$T = N / R$

SHA-1:
$T \approx 10^{15}$ sec $\approx 3.8 \times 10^{7}$ years

SHA-256 / SHA-3:
$T \approx 10^{29}$ sec $\approx 10^{21}$ years

Even the most efficient attack remains infeasible.

Case 3: Computational Power Analysis (CPU, GPU, Supercomputer)

To further evaluate feasibility, we compare real-world computation capabilities.

CPU:
$R \approx 2^{23}$ hashes/sec

GPU:
$R \approx 2^{30}$ hashes/sec

Supercomputer:
$R \approx 2^{50}$ hashes/sec

CPU:
$R \approx 2^{23}$ hashes/sec $\approx 10^{7}$ hashes/sec

GPU:
$R \approx 2^{30}$ hashes/sec $\approx 10^{9}$ hashes/sec

Supercomputer (optimistic upper bound):
$R \approx 2^{50}$ hashes/sec $\approx 10^{15}$ hashes/sec

SHA-256 / SHA-3

Using supercomputer:

$T \approx 10^{38} / 10^{15} = 10^{23}$ sec $\approx 10^{16}$ years

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SHA-1

Using supercomputer:

$T \approx 10^{24} / 10^{15} = 10^9$ sec ≈ 30 years

Interpretation

- Age of Universe ≈ 10^{10} years

Thus:

- SHA-256 / SHA-3 → infeasible even for supercomputers
- SHA-1 → borderline feasible only in theory

Key Insight

Computational power grows linearly, but collision complexity grows exponentially.
Hence, increasing hardware power does not make collision attacks practical.

Case 4: Cryptanalytic Attacks

Cryptanalysis reduces complexity by exploiting weaknesses.

- SHA-1 → reduced complexity → practical attack
- SHA-256 / SHA-3 → no reduction known → remain secure

Final Comparison

Algorithm	Brute Force	Birthday Attack	Cryptanalysis
SHA-1	Impossible	Infeasible	Broken
SHA-256	Impossible	Infeasible	Secure
SHA-3	Impossible	Infeasible	Secure

Final Inference

$2^{(n/2)} >> (\text{hash rate} \times \text{realistic time})$

Therefore, collision search is computationally infeasible for SHA-256 and SHA-3.

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7. SHAttered Attack: Practical Collision in SHA-1

The SHAttered attack demonstrated the first real collision in SHA-1.

Two different PDF files were generated such that:

H(file1) = H(file2), while file1 ≠ file2

How the Attack Works

The attack uses differential cryptanalysis:

1. Select two different prefixes
2. Introduce controlled differences
3. Append blocks to force hash convergence

Computational Effort

≈ 2^63 operations

Equivalent to:

- Thousands of CPU years
- Hundreds of GPU years

Implications

- SHA-1 is no longer secure
- Vulnerable for digital signatures
- Deprecated in modern systems

Comparison

SHA-256 and SHA-3:

- No structural weaknesses
- No reduced complexity attacks
- Remain secure

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8. Practical Perspective and Misconceptions

A common misconception is:

“If no collision is found experimentally, the algorithm is secure.”

Correct interpretation:

Collisions exist, but probability of finding them is negligible.

Even large experiments are insignificant compared to required complexity.

Tools like Hashcat demonstrate high computational speed but are not suitable for collision attacks.

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9. Conclusion

Collisions are inherent in all hash functions due to finite output space. However, security is achieved by making them computationally infeasible to discover.

SHA-1 has been broken due to structural weaknesses, while SHA-256 and SHA-3 remain secure due to their large output size and robust design. The birthday paradox provides the mathematical foundation for this security.

In conclusion, collision resistance ensures that although collisions exist mathematically, discovering them remains beyond any practical computational capability.

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